

A Human-Centric Logarithmic Framework for Enhancing NASA Mars Imagery

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Abstract

NASA's Mars imaging campaigns operate under harsh illumination and atmospheric conditions, producing data with low contrast, sensor noise, dust-induced haze, and wide dynamic range that complicate both human interpretation and automated analysis. Conventional prior-based enhancement methods depend on manually tuned parameters (e.g., noise level, exposure priors, haze assumptions). To address these limitations, we present a human-centric color image enhancement framework for Mars imagery using Logarithmic Color Image Algebra (LCIA), also known as Logarithmic Image Processing (LIP), which models image formation in a manner consistent with the non-linear response of human vision. Instead of standard linear arithmetic, LCIA operates on a logarithmic intensity scale that better matches how observers perceive contrast and brightness under Mars-specific conditions, including strong shadows, backlit slopes, and dust-attenuated skylight. Within this framework, we design operators that (i) reveal fine geological structures in dark or partially shadowed regions without saturating bright regolith, sky, or specular highlights; (ii) preserve chromatic consistency of surface materials and atmospheric features; and (iii) maintain color components (Red, Blue and Green) relationships so that enhanced images remain scientifically better for quantitative analysis. We also use an automatic parameter-adaptation strategy that estimates scene-dependent characteristics directly from the data, reducing manual tuning across Mars datasets. We evaluate the method on real Mars imagery using image quality metrics (contrast enhancement, detail visibility, and denoising), and find that LCIA-based enhancement improves information retention while producing more visually interpretable imagery for mission planning and structural interpretation. The presented pipelines increase the robustness of automated feature detection, segmentation, and edge-based structure analysis in current and future Mars missions.

Introduction

By utilizing logarithmic image processing (LIP) with colored images, we addressed the extreme lighting challenges inherent in Mars datasets, such as dust-induced haze and high-contrast shadows, by modeling image formation through the lens of human visual perception. Unlike traditional linear models that uniformly scale pixel values and often lead to "blown-out" highlights, LCIA operates on a logarithmic intensity scale that mimics the non-linear sensitivity of the human eye to brightness. This mathematical shift allowed us to selectively amplify details in deep shadows, revealing critical geological micro-textures on crater walls and slopes, while simultaneously compressing high-intensity regions to prevent information loss from saturation. By preserving the precise ratios between the Red, Green, and Blue channels, our approach ensures that the resulting imagery is not just visually clearer for human mission planners, but also scientifically consistent for automated feature extraction, such as regolith segmentation and edge-based structural analysis of the Martian surface.

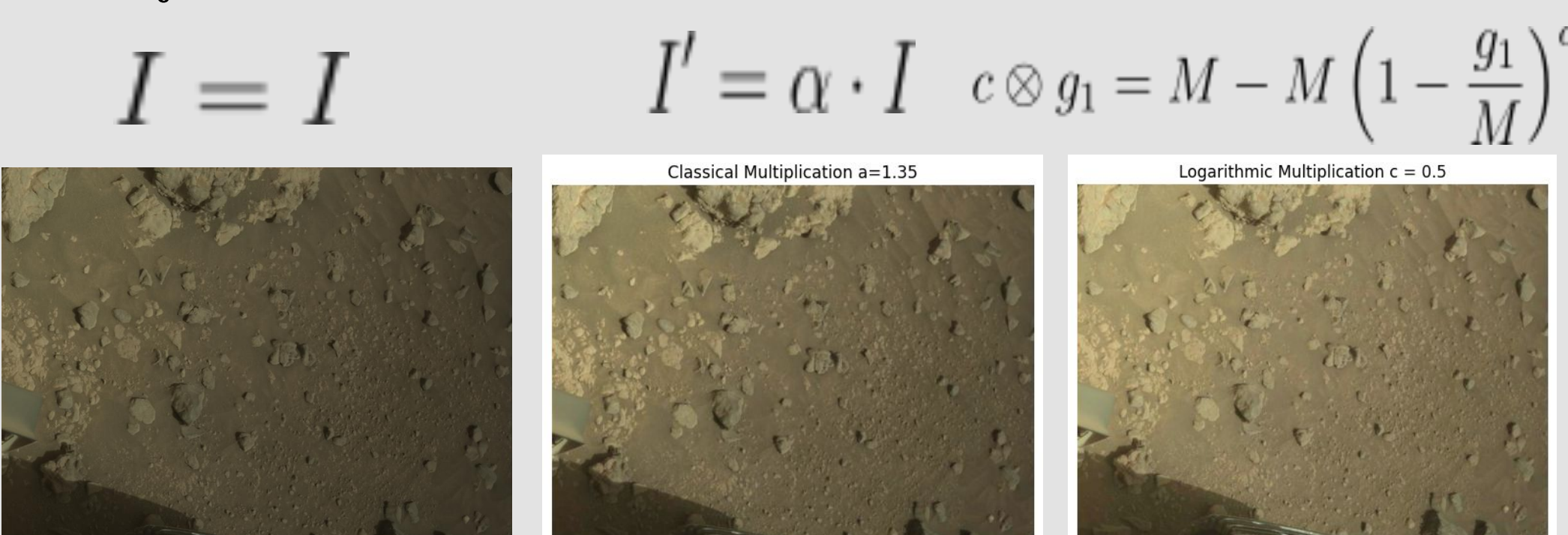
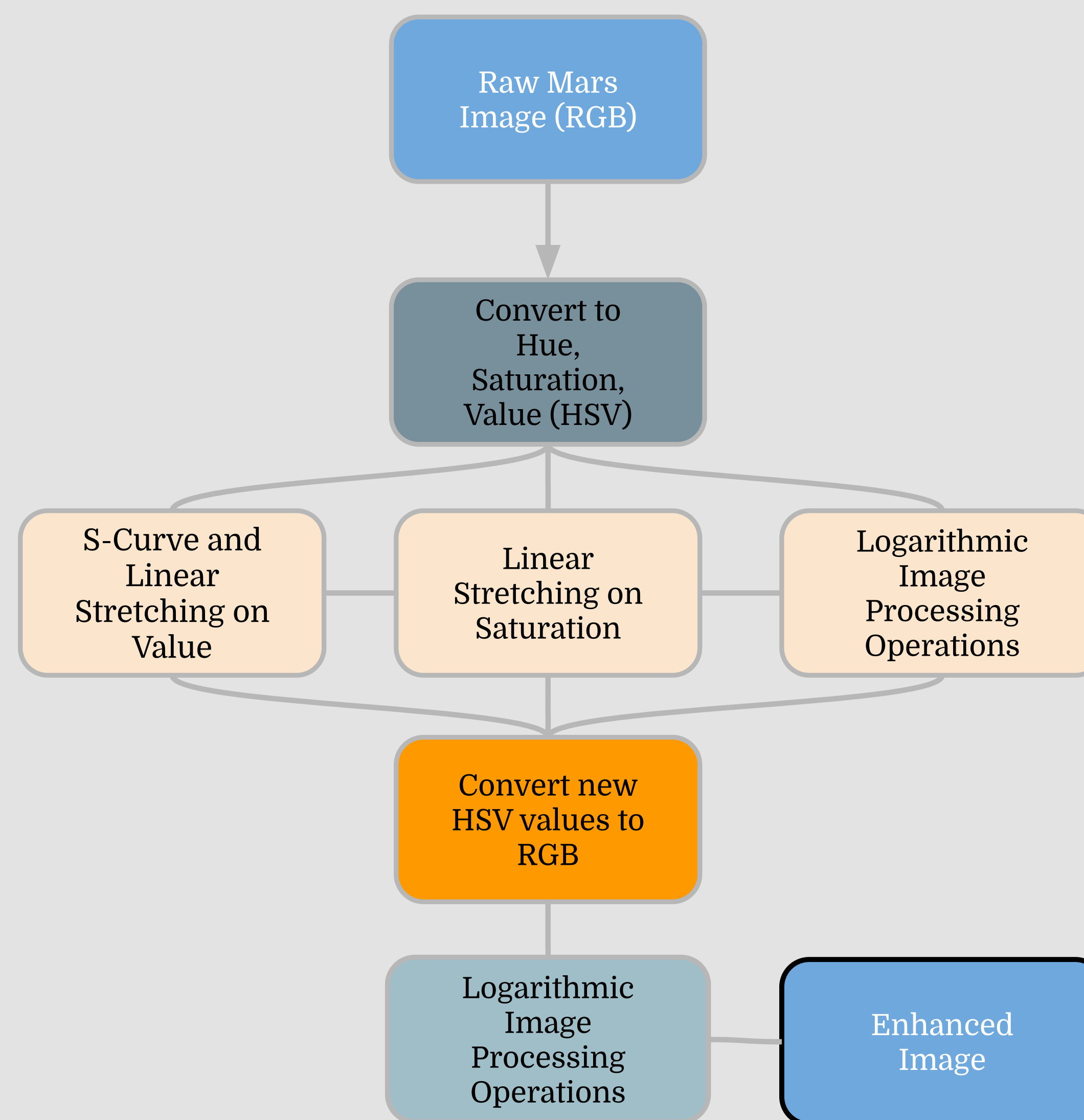


Figure 1: While Classical Multiplication (Left) results in information loss through saturation and shadow masking, the LCIA framework (Right) achieves a human-centric enhancement that reveals critical structural details in both extreme shadows and highlights.

Color Logarithmic Image Processing Framework



Operation	Classical	Logarithmic Image Processing (LIP)
Addition	$a + b$	$a \oplus b = a + b - \frac{ab}{M}$
Subtraction	$a - b$	$a \ominus b = M \frac{a-b}{M-b}$
Multiplication	ab	$a \otimes b = \phi^{-1}(\phi(a)\phi(b))$
LIP Transform	—	$\phi(a) = -M \ln \left(1 - \frac{a}{M}\right)$
Inverse LIP Transform	—	$\phi^{-1}(x) = M \left(1 - \exp\left(-\frac{x}{M}\right)\right)$

S-Curve Contrast Stretch:

$$S(x, y) = \frac{1}{1 + e^{-\alpha(I(x,y) - \beta)}}$$

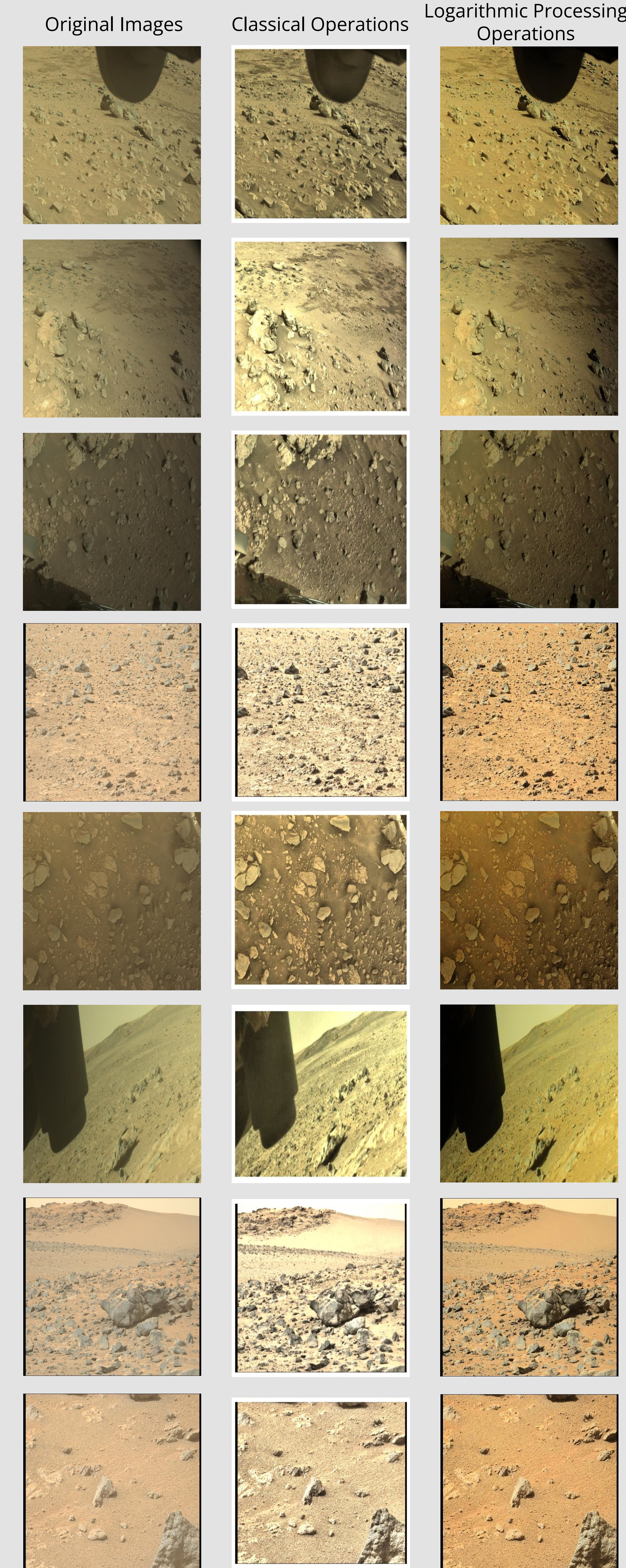
Mars Rover Perseverance Mission Project Dataset



Mars Perseverance Rover Images Dataset:

<https://mars.nasa.gov/mars2020/multimedia/raw-images/>

Results



Conclusion

Our work demonstrates the power of Logarithmic Color Image Algebra (LCIA) by provided a physically realistic and human-centric framework for enhancing Mars imagery under extreme environmental conditions. By operating in a logarithmic intensity domain, the proposed method effectively addressed key challenges in Mars images such as low contrast, dust-induced haze, and high dynamic range, which are not well handled by classical linear techniques.

The LCIA-based pipeline enhanced quantitative and qualitative image quality by revealing fine geological structures in shadowed regions while preventing saturation in high-intensity areas. The integration of adaptive techniques such as CLAHE equalization, adaptive segmentation, and more provides automatic parameter adaptation, which makes the approach robust across the almost one million Mars images collected in the Mars Perseverance dataset.

Our results indicate that logarithmic image processing highly improves visual interpretability when combined with classical image enhancement techniques. The framework we designed is intended to not only improve visual interpretability for human observers but also creates an easy-to-follow algorithm for Mars Perseverance image enhancement.

Future Recommendations

Real-Time Deployment for Rover Systems

- Extend the LCIA framework for real-time onboard processing in Mars rovers (e.g., Perseverance), enabling enhanced visual feedback for navigation and mission planning under dynamic lighting conditions.

Integration with Machine Learning Pipelines

- Incorporate LCIA-enhanced imagery into deep learning models for improved performance in tasks such as terrain classification, rock detection, and anomaly identification.

Adaptive Multi-Scale Enhancement

- Develop multi-scale or region-aware logarithmic operators to selectively enhance features at different spatial resolutions, further improving detail extraction in complex geological scenes.

User-Controlled Visualization Tools

- Design interactive tools that allow mission scientists to adjust enhancement parameters in real time while maintaining the physical constraints of the LIP model.

Sources

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